Centres of Mass

Exam code: 9MA0

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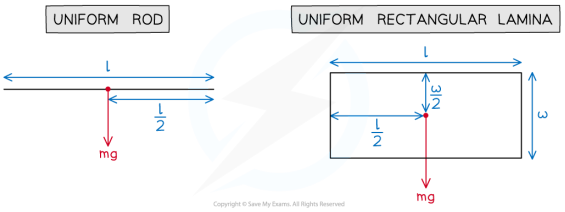
Uniform rods & laminae

What is the centre of mass of an object?

- The centre of mass of an object is **the point at which the weight of the object may be considered to act**
- For a **uniform object** the centre of mass is at the centre of the object where the lines of symmetry intersect
  - For a **uniform rod** this will be at its midpoint
  - For a **uniform rectangular lamina** this will be where the diagonals intersect
- For a **non-uniform** object the centre of mass is not necessarily at the centre of the object



Where is the centre of mass of a uniform rod?

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

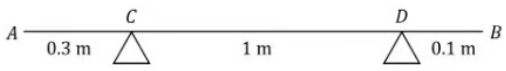
Got it

reaction force which acts **perpendicular to the rod** at that point

- If the rod is **suspended** by **strings** or **cables** then there will be **tensions** in the strings which keep the **rod in place**

Worked Example

A uniform rod AB of mass 20 kg and length 1.4 m rests in equilibrium on two supports at points C and D. AC = 0.3 m and BD = 0.1 m.

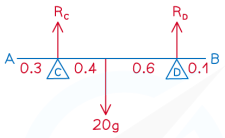


Find the magnitudes of the normal reaction forces at C and D.

Answer:

Let R_C and R_D be the normal reaction forces.
Uniform rod ⇒ centre of mass is 0.7m from A

Step 1: Draw a diagram with all forces and distances



Step 2: Form a force equation

R (↕) $R_C + R_D = 20g$

Step 3: Form a moment equation

M (C) $20g \times 0.4 = R_D \times 1$

$R_D = 8g$

Step 3: Form a moment equation

$R_C + R_D = 20g$

$R_C + 8g = 20g$

$R_C = 12g$

Reaction force at C = 12g N = 120 N (2 sf)
Reaction force at D = 8g N = 78 N (2 sf)

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Examiner Tips and Tricks

- If there are two supports with unknown reaction forces then choosing the pivot to be at one of the supports will help to find the force at the other support. The same method works with strings too.

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Did this video help you? Yes No

Non-uniform rods

How do I find the centre of mass of a non-uniform rod in equilibrium?

- **Step 1**: Label the weight of the rod at a random point on the rod
- **Step 2**: **Call** the perpendicular **distance** between the pivot and the weight x
  - Or any letter of your choice
- **Step 3**: **Form** a **moment equation** using the fact that the rod is in **equilibrium**
- **Step 4**: **Solve** the equation to find x

Worked Example

A non-uniform rod AB of mass 10 kg and length 4 m is suspended by two strings at points A and C with AC = 2.5 m. When the tension in the string at A is 3g N the rod hangs horizontally in equilibrium as shown in the diagram.

Find the distance of the centre of mass of the rod from B.

Answer:

Examiner Tips and Tricks

- Make sure you read carefully which distance you are asked to find.

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